What Separates a Fraudulent Card Transaction from a Normal One?

CSE558 Data Science — Midterm Project Report

Group *Null & Void*
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Code and README: `https://github.com/DarshanBothra/cc-fraud-detection/tree/main/midterm` (`midterm/README.md`; all steps run with `python run_all.py`, fully seeded).

1 Dataset

We use the Kaggle *Credit Card Transactions Fraud Detection Dataset* (`https://www.kaggle.com/datasets/kartik2112/fraud-detection`, CC0), generated with Brandon Harris's Sparkov simulator. It contains card transactions of 999 customers at 693 merchants from 1 Jan 2019 to 31 Dec 2020, split by the author into `fraudTrain.csv` (1,296,675 rows) and `fraudTest.csv` (555,719 rows). We combine both: **1,852,394 rows** with a ground-truth label `is_fraud`; 9,651 are frauds (0.52%), so the classes are heavily imbalanced.

Size rule. Raw: $1{,}852{,}394 \times 23 = 42{,}605{,}062$ cells (22 data columns plus an index). After encoding (Section 2) we keep 14 numeric/derived columns and one-hot encode the 14 merchant categories, giving 28 columns: $1{,}852{,}394 \times 28 = 51{,}867{,}032 \geq 3{,}000{,}000$.

Features and their meaning. `amt`: charge in USD. `category`: merchant type (e.g. `grocery_pos` = in-store groceries, `shopping_net` = online shopping). `trans_date_trans_time`: timestamp. `dob`: cardholder's date of birth (gives age at purchase). `lat/long` and `merch_lat/merch_long`: where the cardholder lives and where the merchant is (gives distance). `city_pop`: population of the cardholder's city. `gender`. **Limitation:** the data is simulated, so our findings describe the generator's fraud profiles, not a real bank; we treat it as a controlled setting for inference.

2 Data preparation

- **Missing values and duplicates.** 0 missing values in all 23 columns; 0 duplicate transaction ids and 0 duplicate rows. Nothing had to be imputed or removed.
- **Derived features.** Hour, night flag (22:00–03:59), day of week and month from the timestamp; age at transaction = (date − `dob`)/365.25; customer–merchant distance by the haversine formula; $\log(1 + \texttt{amt})$; $\log_{10}$ city population; binary gender.
- **Encoding.** One-hot encoding of the 14 categories. We drop names, street, card number, transaction id, merchant name, job, city, zip and state: they identify people or are high-cardinality identifiers with no inferential role.
- **Outliers.** Amounts are extremely right-skewed (median \$47.45, maximum \$28,948.90). We do *not* delete large amounts, because fraud lives in that tail (Section 3); instead we work with $\log(1 + \texttt{amt})$.

Before vs after. The log transform reduces the skewness of amount from 40.81 to −0.30, and the number of points outside $[Q_1 - 1.5\,\mathrm{IQR},\ Q_3 + 1.5\,\mathrm{IQR}]$ from 95,054 to 1,220 (Figure 1). Ages range from 13.9 to 96.2 years and distances from 0.02 to 152.1 km, all plausible, so no further cleaning was needed.

3 Questions, novelty and EDA

Questions. (1) Do frauds involve larger amounts, and by how much? (2) Is fraud a night-time phenomenon? (3) Which merchant categories are riskiest? (4) Are the cardholders hit by fraud older? (5) Does fraud happen farther from the cardholder's home?

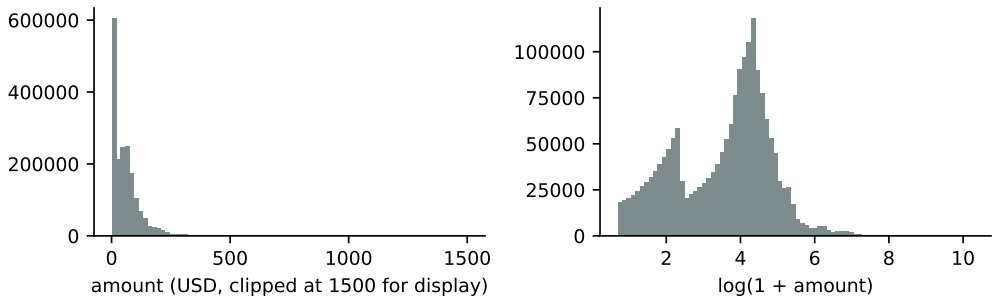


Figure 1: Amount before (left, clipped at $1500 for display) and after (right) the log transform.

Novelty. Most public work on this dataset jumps straight to classifiers and reports accuracy. We instead ask which differences are statistically real *and* large enough to matter, test them on a power-calculated sample rather than on the full data, engineer and test a customer–merchant distance feature, and then check whether each tested signal actually helps a classifier (ablation).

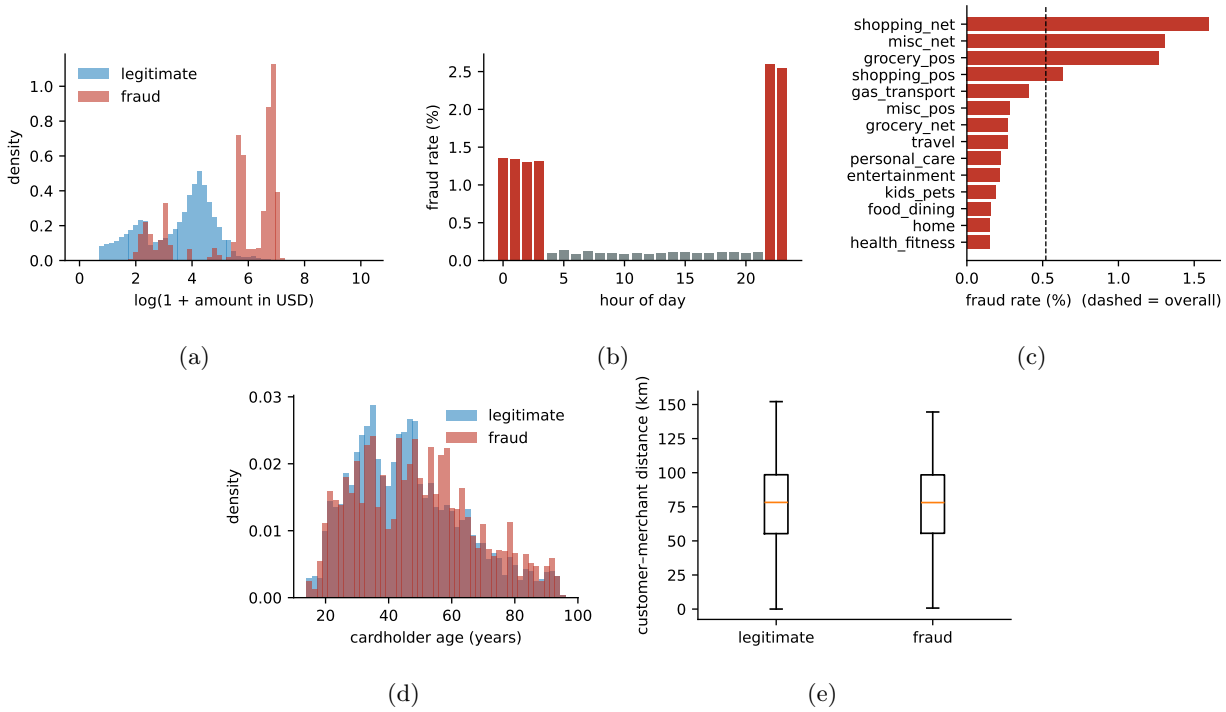


Figure 2: EDA on the full data. (a) Frauds are much larger: median $390 vs $47. (b) The fraud rate is about 2.6% at 22–23h and 1.3% at 0–3h, versus about 0.1% in every other hour (red = night). (c) Online shopping, misc. online and in-store groceries have fraud rates of 1.3–1.6%, about 2.4–3× the overall 0.52%. (d) Fraud victims are slightly older (mean 48.8 vs 46.3 years). (e) Distances look the same (mean 76.3 vs 76.1 km).

4 Statistical inference

4.1 Sampling design

The lecture requirement is to test on a part of the data, not on all of it. On 1.85 million rows every test has so much power that even a trivially small difference gives a tiny p-value, so the p-values would carry no information. We therefore test on a **stratified random sample**. A simple random sample would not work: 4,000 random rows contain only about 21 frauds.

1. **Pilot.** 500 fraud + 500 legitimate rows (seed 7) estimate the standard deviations.
2. **Power calculation** at $\alpha = 0.01$ (Bonferroni for five tests) and power 0.8: $n = 2(z_{1-\alpha} + z_{1-\beta})^2\sigma^2/\Delta^2$ per group for means, and the usual two-proportion formula for Test 2. With the pilot SDs this gives 263 (amount, $\sigma = 1.47$, $\Delta = \ln 1.5$, one-sided), 383 (night share 0.21 → 0.31, one-sided), 810 (age, $\sigma = 17.7$,

$\Delta = 3$ yr) and 786 (distance, $\sigma = 29.0$, $\Delta = 5$ km) per group.

3. **Test sample.** We take **2,000 fraud + 2,000 legitimate** transactions (seed 42), without replacement and disjoint from the pilot, comfortably above every requirement. The sample is representative within each class: means of amount, age, distance and night share are within 6% of the full-data values, and no category share differs by more than 1.1 percentage points. Sample ids are saved in the repository.

Because the classes are sampled in fixed numbers, the sample's 50% fraud rate is artificial. Tests 2 and 3 therefore compare how night-time and categories are distributed *within* frauds and *within* legitimate transactions. Independence is symmetric, so this answers the same question, and odds ratios remain valid under this design (the case–control argument).

Each test uses $\alpha = 0.05$, and we also report whether it survives the Bonferroni level 0.01. For each test we fixed the smallest difference that would matter *before* looking at the sample.

4.2 Test 1: frauds involve larger amounts

$H_0 : \mu_{\text{fraud}} = \mu_{\text{legit}}$ vs $H_1 : \mu_{\text{fraud}} > \mu_{\text{legit}}$ for $\log(1 + \texttt{amt})$, one-sided because the direction was expected. Smallest difference that matters: fraud amounts at least 1.5× larger (geometric means), i.e. $\Delta \geq \ln 1.5 = 0.405$. An F-test rejects equal variances ($F = 1.66$, df $= (1999, 1999)$, $p < 10^{-300}$; SDs 1.63 vs 1.27), so we use **Welch's t-test** rather than the pooled t-test. **Result:** $t = 43.67$, df $= 3763.8$, $p < 10^{-300}$; reject H_0. Difference of log means 2.02 (95% CI $[1.93, 2.11]$), i.e. fraud amounts are **7.5× larger** (95% CI $[6.9, 8.2]$). The whole CI is far above 1.5, so the difference matters. *Other tests:* the z-test gives the same answer at this n but assumes known σ; Mann–Whitney U (no normality assumption) agrees ($p \approx 10^{-285}$).

4.3 Test 2: fraud is more likely at night

$H_0 : P(\text{night} \mid \text{fraud}) = P(\text{night} \mid \text{legit})$ vs H_1: greater (one-sided). Smallest difference that matters: 10 percentage points. **Two-proportion z-test. Result:** 85.4% of frauds vs 23.5% of legitimate transactions are at night; $z = 39.30$, $p < 10^{-300}$; reject H_0. Difference 61.9 pp (95% CI $[59.5, 64.3]$); odds ratio **19.0** ($[16.2, 22.4]$). Far beyond 10 pp, so it matters. *Other tests:* the χ^2 test on the 2×2 table gives $\chi^2 = 1544.6 = z^2$ but is two-sided only; Fisher's exact test agrees but is unnecessary with 2,000 per group.

4.4 Test 3: fraud depends on the merchant category

H_0: category has the same distribution among frauds and legitimate transactions (independence) vs H_1: the distributions differ. χ^2 **test of independence** on the 14×2 table, df $= 13$; the smallest expected count is 49.5, so the approximation is valid. Smallest difference that matters: a category with odds ratio ≥ 2 (fraud vs legitimate). **Result:** $\chi^2 = 742.0$, $p = 3.6 \times 10^{-150}$; reject H_0. Cramér's $V = 0.43$ (strong). Per-category odds ratios (category vs rest): `shopping_net` 3.71 $[3.05, 4.52]$, `grocery_pos` 3.22 $[2.67, 3.88]$, `misc_net` 2.48 $[1.96, 3.14]$; at the other end `home` 0.18 $[0.13, 0.26]$. The CIs of online shopping and in-store groceries lie entirely above 2, so these two matter; misc. online very likely does. *Other tests:* the G-test (likelihood ratio) gives $G = 778.7$, same conclusion; Fisher's exact test is impractical for a 14×2 table this size.

4.5 Test 4: fraud victims differ in age

$H_0 : \mu_{\text{fraud}} = \mu_{\text{legit}}$ (age) vs H_1: they differ (two-sided; no prior direction). Smallest difference that matters: 3 years. **Welch's t-test. Result:** $t = 4.40$, df $= 3994.2$, $p = 1.1 \times 10^{-5}$; reject H_0. Fraud victims are **2.6 years older** (95% CI $[1.4, 3.7]$). The CI contains 3, so we cannot say the difference matters; an equivalence test (TOST, ±3 years) does not establish that it is negligible either ($p = 0.23$). *Conclusion: real but small, practical relevance unclear. Other tests:* Mann–Whitney agrees ($p = 4 \times 10^{-6}$); Kolmogorov–Smirnov ($D = 0.107$, $p = 2 \times 10^{-10}$) shows the age *distributions* differ in shape, not just in mean (visible in Figure 2d).

4.6 Test 5: does fraud happen farther from home?

$H_0 : \mu_{\text{fraud}} = \mu_{\text{legit}}$ (distance in km) vs H_1: they differ (two-sided). Smallest difference that matters: 5 km. **Welch's t-test. Result:** $t = 1.94$, df $= 3998.0$, $p = 0.053$; we **fail to reject** H_0. Difference 1.8 km (95% CI $[-0.02, 3.54]$). Failing to reject does not prove equality, so we add an equivalence test: TOST with bounds ±5 km rejects non-equivalence ($p = 0.0002$). Since the sample was powered to detect 5 km, we conclude that **any distance difference is practically negligible**. This suggests the simulator places merchants independently of whether a transaction is fraudulent. *Other tests:* Mann–Whitney is borderline ($p = 0.046$, not significant at 0.01); KS $p = 0.064$.

4.7 Summary and robustness

Test	Method	Statistic	p	Estimate (95% CI)	Reject at 0.01?	Matters?
1 Amount	Welch t (1-sided)	$t = 43.67$	$p < 10^{-300}$	ratio 7.5 [6.9, 8.2]	yes	yes
2 Night	2-prop. z (1-sided)	$z = 39.30$	$p < 10^{-300}$	+61.9 pp [59.5, 64.3]	yes	yes
3 Category	χ^2 indep., df 13	$\chi^2 = 742.0$	10^{-150}	$V = 0.43$; OR up to 3.7	yes	yes
4 Age	Welch t (2-sided)	$t = 4.40$	10^{-5}	+2.6 yr [1.4, 3.7]	yes	unclear
5 Distance	Welch t (2-sided)	$t = 1.94$	0.053	+1.8 km [−0.02, 3.5]	no	no (TOST)

Table 1: Hypothesis tests on the stratified sample (2,000 fraud + 2,000 legitimate).

Repeating all five tests on five further independent samples (seeds 1001–1005) gives the same decision at $\alpha = 0.01$ every time. As an extra, a χ^2 goodness-of-fit test shows that frauds are not spread across the days of the week like legitimate transactions ($\chi^2 = 88.5$, df $= 6$, $p = 6 \times 10^{-17}$; fraud rates are highest Wednesday–Friday).

5 Using the signals: binary classification

Since the data is labelled, we fit supervised binary classifiers and check whether the signals found above are the ones a model needs. We train on `fraudTrain.csv` (later split 80/20 into training and validation) and evaluate once on `fraudTest.csv` (555,719 rows, 2,145 frauds; later months, so no leakage). Features: log amount, hour, night flag, day of week, age, gender, log city population, distance and the 14 category dummies. All modelling choices and the decision threshold (maximum F1) were made on the validation split only. Class weighting lowered validation PR-AUC for both models, so we handle imbalance with the threshold instead. We report precision, recall, F1 and PR-AUC, not accuracy: predicting "legitimate" for everything already scores 99.6%.

Model (test set)	Precision	Recall	F1	PR-AUC	ROC-AUC
Logistic regression, linear terms	0.32	0.58	0.41	0.39	0.90
Logistic regression, spline terms (amount, hour, age)	0.64	0.70	0.67	0.70	0.95
Decision tree (depth 12, min. leaf 20)	**0.87**	**0.78**	**0.82**	**0.85**	**0.98**

Table 2: Classifier performance on the held-out test file.

Model comparison. Plain logistic regression is weak because the effect of amount is not monotone: the largest spline coefficients for amount change sign across its range (−11.8, +7.2, +7.3). Adding spline terms lifts PR-AUC from 0.39 to 0.70. The decision tree captures these interactions directly and is best. **McNemar's test** on the test-set predictions (H_0: both models have the same error rate): the tree is right where logistic regression is wrong 944 times, the reverse 180 times; $\chi^2 = 517.9$, $p \approx 10^{-114}$. A paired bootstrap (200 resamples) gives the tree an F1 advantage of $[0.14, 0.17]$ and a PR-AUC advantage of $[0.13, 0.16]$ (95% CIs).

Do the tested signals matter in the model? We drop each tested feature group, refit, and measure the fall in test PR-AUC (95% bootstrap CI, 100 resamples):

Feature group dropped	Fall in PR-AUC: spline LR	Fall in PR-AUC: tree	Hypothesis test said
Amount (Test 1)	0.48 [0.46, 0.50]	0.61 [0.59, 0.63]	large
Time of day (Test 2)	0.11 [0.09, 0.13]	0.11 [0.10, 0.13]	large
Category (Test 3)	0.32 [0.29, 0.33]	0.21 [0.19, 0.23]	large
Age (Test 4)	0.007 [0.005, 0.010]	0.04 [0.02, 0.05]	small, unclear
Distance (Test 5)	0.000 [−0.0001, 0.0001]	0.000 [−0.003, 0.003]	negligible

Table 3: Ablation: drop in test PR-AUC when a feature group is removed (CI of the drop).

The ablation lines up with the inference. The three features with large, practically important differences (amount, category, time of day) are the ones the models cannot do without. Age, statistically significant but small, adds little. Distance, shown by TOST to be negligible, adds nothing. The tree's feature importances tell the same story (log amount 0.31, `grocery_pos` 0.25, night 0.14).

6 Conclusion and limitations

Frauds in this data are **about 7.5× larger**, **overwhelmingly at night** (85% vs 23%, odds ratio 19), and **concentrated in online shopping and in-store groceries** (odds ratios 3.2–3.7). Fraud victims are slightly older (2.6 years), a real difference whose practical relevance our sample cannot settle. Fraud does *not* happen farther from home: an equivalence test shows any difference is under 5 km. These conclusions come from a power-calculated stratified sample, are stable across five further samples, and are confirmed by an ablation of two classifiers, where a depth-12 decision tree reaches PR-AUC 0.85 on unseen later months.

Limitations. The data is simulated, so effects reflect the generator's design. The tests use one sample (although five replications agree). Five tests are run, which we handle with Bonferroni. Stratified sampling cannot estimate absolute fraud rates, which we report only descriptively from the full data. Customers repeat across rows, so observations are not fully independent.